ChatGPT

Comprehensive Review of *"Predictive AI for Active Aerodynamics Efficiency and Driver Anomaly Analysis in Formula 1 Racing"*

PHASE 1: Paper Understanding

1. **Research Domain:** The paper lies at the intersection of *Formula 1 motorsport analytics* and *machine learning*. It specifically addresses **active aerodynamic systems** mandated in F1 (the new Z-Mode/X-Mode wing configurations in the 2026 regulations 【16†L100-L103】) and the analysis of *driver telemetry data* for anomaly detection. In broader terms, it falls under AI for automotive/motorsport systems and time-series prediction.

2. **Research Problem:** The authors tackle two related problems: (a) **Active Aerodynamics Mode Prediction:** how to anticipate the switching between high-downforce (Z-Mode) and low-drag (X-Mode) wing configurations in F1 cars one second before the switch, in order to overcome the physical actuation latency. (b) **Driver Anomaly Detection:** identifying unusual or extreme driving events (e.g. hard braking) from telemetry, which may indicate performance anomalies or hazards. The core challenge is to build predictive models from race telemetry that generalize across different circuits ("cross-circuit generalization") without retraining.

3. **Proposed Solution:** A **two-phase AI framework** is proposed:

4. *Phase 1:* Unsupervised anomaly detection combined with supervised state classification. They apply Isolation Forest and K-Means clustering to find anomalous telemetry points, then train classifiers (logistic regression and random forest) on current aerodynamic mode labels. The goal is to validate the approach on one race's data (Japanese GP) and understand feature importance.

5. *Phase 2:* Temporal predictive modeling across circuits. They train a variety of sequence classifiers (from simple logistic regression on the most recent frame up to deep models like CNN, LSTM, GRU, TCN, and Transformers) to predict X-Mode vs Z-Mode one second ahead. Evaluation uses a **Leave-One-Circuit-Out (LOCO)** protocol across four circuits, to test cross-track generalization. An integrated gradients interpretability analysis is also applied to diagnose model failures.

6. **Main Contributions (claimed):** The authors explicitly list several contributions (in abstract and introduction), including: (i) A *two-phase evaluation framework* (unsupervised anomaly + supervised prediction). (ii) A **circuit-adaptive performance metric** and LOCO evaluation across four F1 circuits. (iii) An *empirical demonstration* that a simple model (logistic regression) outperforms deep sequential models for the 1-second-ahead prediction. (iv) Use of **integrated gradients** analysis to explain a GRU model's failure (speed memorization issue). (v) Significance testing (bootstrap confidence intervals) to confirm that the simple model's advantage is statistically significant.

7. **Key Methodologies:** The paper employs standard and advanced ML techniques: *Isolation Forest* for unsupervised anomaly detection【18†L235-L243】, *K-Means* clustering on standardized features, *Permutation Importance* with bootstrap for feature importance. For prediction, they use logistic regression and random forest as baseline classifiers, plus a suite of *deep learning* sequence models: 1D CNN, temporal convolutional networks (TCN)【10†L0-L4】, LSTM, GRU【24†L169-L177】【25†L169-L177】, and a causal (masked) Transformer【23†L26-L34】. They apply *focal loss*【26†L23-L27】 and AdamW optimizer for training, with early stopping on validation AUC. Integrated Gradients【7†L23-L27】 is used to interpret feature attributions in the trained GRU. Overall, the methods are sound and well-known in ML.

8. **Datasets Used:** The authors use official F1 "FastF1" telemetry data from four Grand Prix races (one per circuit): Suzuka 2026 (63,673 frames), Monaco 2025 (39,881), Monza 2025 (31,098), and Silverstone 2025 (25,060), for a total of 159,712 frames. Data were sampled at 10 Hz. For Phase 1 (anomaly + classification), only Suzuka 2026 is used. Phase 2 uses all four (with LOCO splitting). Features include speed, RPM, gear, throttle, brake, GPS coordinates, lap distance, plus derived kinetic energy and longitudinal force (total 12 features).

9. **Evaluation Metrics:** The main metrics are *Area Under the ROC Curve (AUC-ROC)*, overall *F1-Macro*, and *F1-X-mode* (F1 score for the X-Mode class). They also use bootstrap confidence intervals for AUC differences and permutation tests for statistical significance. These choices are appropriate for a binary classification problem with imbalance (X-Mode is about one-third of data).

10. **Major Findings:**

11. **Phase 1:** The random forest (RF) classifier achieves *perfect detection* of current mode (AUC=1.000). The largest anomalies (Isolation Forest outliers) coincide with heavy braking zones. Permutation importance shows *gear* is the single most discriminative feature (dropping gear prediction reduces AUC by 0.167).

12. **Phase 2:** Across LOCO experiments, simple models excel. Logistic regression (LR) on instantaneous features achieves mean AUC≈0.963, outperforming all deep models. Notably, a 2-layer GRU fails on Monaco data (AUC≈0.728, F1-X≈0.005); integrated gradients reveal that the GRU had overfit to absolute speed values (circuit-specific scales), causing the failure. In contrast, relying on instantaneous threshold-features (speed ≥ 290km/h and gear≥6) yields robust generalization. Bootstrap CIs show the LR's superiority is statistically significant (95% CI excludes zero). The authors conclude that "instantaneous physics features generalize best across circuits" for this task.

Assessment of Claimed Contributions: All stated contributions are indeed present in the paper. The two-phase framework is clearly executed, the LOCO protocol is followed with four circuits, and the surprising empirical result (LR>AUC=0.963 vs. deep models) is thoroughly documented. The integrated gradients analysis and bootstrap testing are also actually performed as claimed. In summary, the contributions listed by the authors are **genuinely supported** by the experiments and analysis in the paper.

PHASE 2: Section-by-Section Review

Title

- **Clarity:** The title is descriptive but somewhat **long and complex**. It accurately mentions “Predictive AI,” “Active Aerodynamics Efficiency,” and “Driver Anomaly Analysis” in F1. However, the connection between “active aerodynamics” and “driver anomaly” is not immediately obvious from the title alone; they appear as two separate topics joined by “and.”
- **Relevance:** It contains relevant keywords (active aerodynamics, anomaly analysis, F1), which is good for indexing. It reflects both the predictive and analytical aspects of the paper.
- **Technical Accuracy:** The title correctly identifies the domain (Formula 1) and the high-level tasks (predictive modeling for aerodynamics and anomaly analysis).
- **Attractiveness:** The phrasing is somewhat awkward. The use of “Predictive AI” is catchy but vague, and “Efficiency and Driver Anomaly Analysis” is somewhat jargony. It reads more like a list of buzzwords than a cohesive phrase.

Suggestions: Consider simplifying. For example, “Predictive Modeling of Active Aerodynamic Mode and Telemetry Anomalies in Formula 1” (or similar). This would directly convey the forecasting task (mode changes) and anomaly detection in one coherent phrase. Reducing length and using clear terms (e.g. “telemetry anomalies” instead of “driver anomaly analysis”) would improve clarity and impact.

Abstract

- **Motivation & Problem Statement:** The abstract starts by motivating active aero in F1 and the issue of latency. This is good. It clearly states the problem of predicting mode changes to reduce latency.
- **Methodology:** It outlines the two-phase approach succinctly (unsupervised anomaly detection, supervised classification, cross-circuit predictive modeling with various models).
- **Results:** Key results are given (RF AUC=1.000, LR outperforms deep models with AUC≈0.963, GRU failure at Monaco, gear importance, etc.). This is excellent – it quantifies findings.
- **Contributions:** The abstract mentions a “circuit-adaptive metric,” LOCO evaluation, model comparisons, and interpretability findings. These cover the contributions.
- **Keywords:** Implicitly includes relevant concepts (F1, active aero, anomaly detection, predictive models, cross-circuit). Could explicitly list keywords.

Weaknesses:

- **Missing elements:** The abstract is very dense, perhaps too detailed. It could state upfront that the study covers both *anomaly detection* and *mode prediction*, to make the two-phase nature clearer.
- **Unclear Wording:** Some phrases are clunky (e.g. “unconstrained physics features” is confusing). The sentence about LOCO with horizon H=10 is a bit jargon-heavy.
- **Generic Statements:** It ends with a somewhat generic conclusion (“Instantaneous features generalise best across circuits”). More specific mention of why (thresholding is circuit-invariant) would be stronger.

Suggested Improved Abstract (IEEE-style):
“**Abstract—** The 2026 F1 regulations introduce active front/rear wings switching between high-downforce and low-drag modes. To overcome actuation delays, we propose a predictive AI framework that anticipates mode changes 1 s in advance. In Phase I, we apply unsupervised outlier detection (Isolation Forest) and

clustering on telemetry from the 2026 Japanese GP to identify anomalous braking events, and train classifiers (logistic regression, random forest) for current mode detection (achieving AUC = 1.000). In Phase II, we evaluate 8 models (from instantaneous logistic regression to deep CNN/LSTM/TCN/Transformer) across four circuits using a Leave-One-Circuit-Out protocol. We find that a simple instantaneous logistic model (AUC≈0.963±0.021) outperforms all deep sequence models (bootstrap-CI, $p \ll 0.05$). Integrated Gradients analysis reveals that the GRU's poor Monaco performance (AUC≈0.73) stems from overfitting speed distributions, whereas threshold-based features (speed ≥290 km/h and gear ≥6) generalize. Our results highlight that physics-driven threshold features yield robust cross-circuit predictions of active-aero mode."

Introduction

- **Problem Motivation:** The introduction effectively motivates the need for predictive active aero in F1 (latency of 100–300 ms causes significant track inefficiencies at 350+ km/h). This establishes technical importance. However, it could strengthen the *broader impact* (e.g. how improving aero switching could affect lap times or fuel efficiency).
- **Literature Gap:** It notes prior work on F1 telemetry (e.g. tire degradation, driver classification) and correctly claims that *anticipatory active aero prediction* is novel. This gap statement is clear, though it would help to cite any directly related predictive control work or general *domain adaptation* research to show novelty (cf. domain adaptation theory 【6†L51-L59】).
- **Significance:** The introduction ties the problem to upcoming 2026 rules, which is timely and important for the field. It mentions expected performance improvements. It convincingly justifies the research question.
- **Research Objectives:** The goals are implied (two-phase predictive analysis), but the text could explicitly state "We aim to (1) detect anomalies in telemetry, and (2) predict mode switches across tracks." Currently it's woven into narrative rather than listed.
- **Contributions Summary:** The listed contributions are relevant (LOCO protocol, ML models, interpretability). They are clearly itemized.

Weaknesses:

- **Missing Citations:** The statement about 2026 active aero modes could cite an FIA source or technical news 【16†L100-L103】. The claim of "no prior work" might be backed by a general reference on domain shift (e.g. Ganin's domain-adaptation theory 【6†L51-L59】) or recent anomaly surveys.
- **Flow/Poor Transitions:** The introduction abruptly jumps between F1 facts, then to literature, then contributions. A clearer outline of the paper at the end would help flow. For example, after listing contributions, briefly outline the section structure.
- **Unsupported Claims:** The effectiveness of predictive switching is asserted but not quantified (how many meters of delay saved?). It might oversell by implying guaranteed improvement without evidence.

Suggestions: Add one or two citations for background (e.g. FIA regulations 【16†L100-L103】). Clarify objectives in a list. Ensure each claim is modestly phrased (e.g. "can potentially improve lap efficiency"). Smooth transitions (e.g. "To this end, we propose…").

Literature Review / Related Work

- **Coverage:** The related work is split into two parts. Part A (Aerodynamics) cites classic sources [7] and modern predictive modeling in motorsport [8][9]. Part B (Anomaly Detection) covers general anomaly methods [10][30] and some automotive telemetry [11][12][13]. This is a reasonable spread.
- **Missing Seminal Works:** One might expect mention of *cross-domain learning* or *transfer learning* for time series, since the paper focuses on generalization across tracks. Including references on domain adaptation or multi-domain time series (e.g. Ganin [6] or recent domain-generalization studies) would strengthen context. Also, any prior work on "leave-one-track-out" evaluations in sports analytics would be relevant if it exists.

- **Recent References (Last 3–5 years):** The authors cite up to 2025 in anomaly detection [12]. However, the aerodynamics section relies on a 2006 book [7] and cites mostly 2020–21 work. There are likely newer papers on ML-driven aerodynamic optimization or vehicle dynamics in the last few years. For example, one could cite recent reviews of *active aerodynamic systems in vehicles* or *ML in motorsport engineering*.

- **Comparison Quality:** The related work descriptions are brief. In Aerodynamics, the cited works [7][8][9] are only tangentially related (DRS/prediction of DRS, wing configuration optimization). A deeper comparison of how those problems differ from active mode prediction would be helpful. In Anomaly Detection, they cover good ground but mostly focus on consumer vehicles or general methods; mentioning F1-specific challenges (e.g. extreme class imbalance, sensor noise) would contextualize their approach.

- **Research Gap Justification:** The review does justify the gap by omission (no existing anticipatory mode prediction papers are cited). But it could be more explicit in summarizing that gap (e.g. "Unlike prior work which is reactive or single-track, we study predictive, cross-track models").

Suggestions: Include domain adaptation literature【6†L51-L59】 as a reference for the cross-circuit challenge. Add a recent survey on anomaly detection for time series or motorsport (e.g. Ruff *et al.* on anomaly detection【8†L209-L212】) to bolster section B. Compare at least one work in detail: e.g. how is X-Mode prediction different from DRS prediction [8] or anomaly detection [13]? Finally, remove any tangential citations (e.g. outdated or irrelevant as noted later).

Methodology / Proposed Framework

- **Technical Correctness:** The methods are generally correct. The Isolation Forest scoring formula (Eq. 5) is right. The description of neural architectures (LSTM, GRU, TCN, Transformer) matches standard designs. The use of focal loss and AdamW is appropriate for imbalanced time-series classification.
- **Completeness:** Mostly complete. They specify hyperparameters (number of trees, depth, learning rates). The sliding window (5 s history, 50 frames) is clearly defined. The LOCO evaluation is implicit. However, some details are missing:
- The **K-Means clustering** step in Phase 1 is not described in Section IV.A of the paper (we only see "K-Means" in results). It should be explained in Methodology how clustering was used (e.g. k=4 found by elbow method) – in fact the results mention k=4, but the method text seems to skip explaining this step.

- **Data splitting:** They mention an 80/20 split for classification, but later use LOCO for prediction. It would help to clarify how validation sets are chosen in each phase. For Phase II, it's inferred they train on 3 circuits and test on 1, but an explicit statement of the LOCO procedure would aid reproducibility.

- **Reproducibility:** The paper lists most model hyperparameters and training procedures, which is commendable. One weakness is that random seeds or software/library versions (e.g. PyTorch, FastF1) are not mentioned. Also, hardware details (GPU type, training time) are absent. Finally, the exact train/validation split method (stratified shuffle vs. by laps) could be more precise.

- **Algorithm Explanations:** Equations are explained (Isolation Forest score). The logic of classification/regression is straightforward. One small flaw: Eq.(5) has formatting issues (the exponent and division layout is not clear in text, likely a typesetting glitch). They should ensure the math is readable.

- **Unsupported Assumptions:** An assumption is that the X/Z mode label is entirely determined by speed and gear (indeed that is by rule【16†L100-L103】). This justifies the high performance of LR. It would be good to explicitly mention this assumption in methods. Another assumption is that anomalies can be captured by Isolation Forest on standardized features – generally valid, but the choice of contamination=0.05 seems arbitrary (though they say results were stable across 0.02–0.10).

Suggestions: Add a brief description of the K-Means clustering step and its purpose (e.g. grouping telemetry patterns). Clearly outline the LOCO evaluation setup (e.g. "each circuit is held out as a test set in turn"). Include any random seed setting or platform info for reproducibility. Fix any equation formatting (Equation 5 is currently hard to read). Ensure it's clear that the X/Z label is a deterministic threshold function – either here or in Data section.

Experimental Setup

- **Dataset Quality:** The telemetry is high-quality professional data. The authors explain how it was acquired (FastF1) and the features used (12 total, including physics-derived ones). It might be useful to mention race conditions (e.g. dry vs wet, which driver/car the data comes from) – variability here could affect generalization. They should clarify if data from multiple drivers or just one car was used.
- **Train-Test Split:** The 80/20 stratified split for Phase 1 is standard, but more important is the LOCO split in Phase 2. They should explicitly describe in text that "for Phase II, we train on three circuits and test on the fourth in each fold." Also, how is the 80/20 split applied within circuits – is it across laps or frames?
- **Baselines:** They include reasonable baselines: instant LR/RF and a "RF-lag" which uses 5-second history without temporal modeling. This is good. One omission: **no naive baseline** (e.g. always predict Z-mode or threshold heuristic) is shown. Including a trivial rule-based predictor (e.g. gear≥6) would highlight how much ML adds.
- **Hyperparameters:** Hyperparameters are mostly listed (LR C=1, RF depth, dropout, etc). One table (Table IV) is presumably given. However, it would be helpful to know how these were chosen (e.g. via grid search or default).
- **Hardware/Software:** The paper does not mention what hardware (GPU) or frameworks (TensorFlow/PyTorch) were used. These are minor but recommended for reproducibility.

- **Evaluation Procedure:** The authors explain which metrics they compute (AUC, F1). However, details like threshold selection (for F1) are not given explicitly. For significance, they use bootstrap CIs, which is good. Possibly mention number of bootstrap samples.

Suggestions: Clarify the source of data (e.g. driver/team), split method (per-lap vs. random frames), and note any data preprocessing (normalization used?). Include simple baseline (e.g. "always predict X-mode if speed>290km/h") for reference. List hardware and software versions. Specify any hyperparameter tuning approach. If class weights or oversampling were tried as alternatives to focal loss, mention that choice.

Results and Discussion

- **Tables & Figures:** The figures (ROC curves, bar charts of AUC, F1 scores) are informative but a bit small in the PDF. The captions are descriptive. Table V/VI (not shown here) presumably list metrics. All key results (means, std or CI) are reported, which is good. However, it would help if confusion matrices or precision/recall were shown for the best/worst models to give more insight into trade-offs.
- **Statistical Validity:** The authors use 95% bootstrap confidence intervals to test differences, which strengthens the claims. They mention "CI excludes zero" for performance gap – this is rigorous. They could complement this with a formal test (e.g. DeLong's test for AUC differences) to further solidify significance. Also, reporting p-values or exact CI values would help the reader gauge effect sizes.

- **Comparative Analysis:** The comparative results are clearly discussed: logistic regression wins by a margin, deep models lag. The surprising GRU failure at Monaco is well highlighted. One concern: they only test one horizon (H=1 s). It would be interesting to see how performance changes for shorter or longer horizons. They also do not compare to any non-learning rule (see above), so it's hard to know how much ML outperforms a simple heuristic.

- **Conclusions vs Results:** The claims in the discussion match the results tables. The claim "LR significantly outperforms deep models" is backed by data. They do not appear to overstate; the failure at Monaco is honestly reported.

Weaknesses / Missing:

- **Ablations:** Missing ablation studies. For example, what happens if they remove one feature (like gear) entirely from the best model (beyond permutation)? What if horizon H is varied? Also, the effect of focal loss vs. standard cross-entropy is not isolated.
- **Benchmarking:** Aside from their models, there is no external benchmark or reference (possibly because none exists). However, a simple persistence or rule-based baseline (as mentioned) would quantify the ML benefit.
- **Potential Objections:** A reviewer might ask whether the results could be due to overfitting on the limited data. The LOCO scheme counters this. Another concern is model complexity vs data size; deep models may not have enough data to outperform shallow ones, which the authors partially acknowledge with early stopping. It could be pointed out that more data (more races) might help.
- **Visualization:** It might help to visualize the Integrated Gradients weights (speed vs time) to support the textual diagnosis, but this might be beyond scope.

Suggestions: Perform additional ablations (vary horizon, remove features, try simpler rule). Include or mention a simple heuristic baseline for context. If space permits, add precision/recall or confusion matrices.

Ensure all claims about significance are accompanied by specific CI or p-values. Possibly discuss why deep models underperform (lack of data or model mis-specification) in more detail.

Conclusion

- **Alignment with Findings:** The conclusion accurately summarizes the work: it restates the two-phase study and highlights the main outcomes (RF detection, LR wins, GRU issue). It does not introduce new claims.
- **Contribution Summary:** It recaps the contributions clearly, matching what was set out. It emphasizes the practical insight (instantaneous features generalize best).
- **Future Work:** The paper has a separate "Future Work" section that mentions "Domain-Relative Normalization" (using relative speed) – this is a good specific suggestion. The conclusion itself does not list future work (it was probably merged with section VIII).

Weaknesses: The conclusion reads a bit like an expanded abstract rather than reflecting critically on limitations or future directions. It could do more to contextualize impact (e.g. "This suggests that simple physics-based thresholds might suffice in practice【6†L51-L59】"). Also, it might overgeneralize by implying "integrated gradients revealed failure modes" – this was only demonstrated for one model.

Suggestions: The conclusion should briefly note any limitations (e.g. limited to certain tracks or sensors) and explicitly connect to broader implications (e.g. for F1 engineers). One could suggest adding that domain adaptation techniques (as in Ganin et al. 【6†L51-L59】) might improve cross-track learning, hinting at future research. Also, ensure the final sentence highlights the novelty or lesson: e.g., "In short, domain-invariant features based on F1 physics yield robust predictions."

PHASE 3: Citation & Reference Audit

Contextual Relevance: Overall, citations are placed mostly appropriately. However, a few statements could use support from the literature. For example, the challenge of cross-track generalization is not cited – here one could reference domain-adaptation theory (Ganin *et al.*). The paper does cite methodological sources (e.g., [26], [27]) correctly.

Placement & Support: Most references back up background or methods. One oddity is that reference [30] (Chandola) appears inserted in text as "[30]. Isolation Forest [10] has been applied…", which seems like a formatting error (the number 30 probably belonged to the end of the previous sentence). Otherwise, references like [10] for Isolation Forest and [26], [27] for focal loss/IG are well-placed. The text "Causal LSTM [24]" properly cites Hochreiter's LSTM [24] (1997).

Unnecessary Citations: The reference list includes a few citations that are tangential to the paper's focus (discussed below). Some of these are not actually cited in the text at all. For example, [21] (Kostas *et al.*, EEG data) and [22] (Zhang *et al.*, machinery fault) have no clear relevance to F1 or time-series prediction. These should likely be removed.

Missing Citations:

- The statement "2016 regulations mandated active aero" (if present) should cite the actual FIA document or credible report【16†L100-L103】.

- The concept of cross-domain generalization (multiple circuits) has no citation; a domain adaptation or domain generalization reference (e.g. Ganin *et al.*) would be appropriate【6†L51-L59】.
- The use of permutation importance is not cited (this is a known technique, but could cite Breiman or a recent primer).
- In “Related Work,” the review of anomaly detection could cite a recent survey (e.g. Ruff *et al.*, 2021)【8†L209-L212】 for up-to-date context.

Citation Table:

Section	Statement	Why Citation Needed
Introduction	“2026 regulations introduce active aerodynamic Z/X modes.”	To verify new rule and define Z/X modes (e.g. FIA regulation or technical summary【16†L100-L103】).
Introduction	“No prior work on predictive active-aero switching.”	To substantiate novelty or compare to related predictive control literature.
Related Work	(Domain generalization gap)	No references on domain adaptation or cross-track ML; see Ganin et al.【6†L51-L59】.
Methodology	(Permutation importance)	Could cite a source on permutation feature importance (e.g. Breiman’s work).
Conclusion	(Generalization of physics features)	A reference on domain-invariant features would add support (e.g. Ganin【6†L51-L59】 again).

Weak References: The following references are weakly connected or outdated in the current context:

Reference	Issue	Suggested Replacement / Improvement
[21] Kostas *et al.* (2021) BENDR (EEG data)	Irrelevant domain (EEG); not applicable to motorsport telemetry.	Remove or replace with domain-related work (e.g. Bai *et al.* on TCN vs RNN【10†L0-L4】).
[22] Zhang *et al.* (2021) Machinery fault	Irrelevant; focuses on vibration fault detection in machinery.	Replace with an anomaly detection review in automotive contexts (e.g. Ruff *et al.*, 2021【8†L209-L212】).
[23] Ordóñez & Roggen (2016) Human Activity Recog.	Dated (2016) and not directly on vehicular data.	Use a more recent time-series classification review or Bai *et al.* (2018) comparison【10†L0-L4】.
[4] Peterson (2018) motorsport RNN for sprint	Only peripherally related to predicting car dynamics, domain mismatch.	Possibly replace with an F1 telemetry study or skip if out-of-scope.
[14] Bunker & Thabtah (2019) sports ML survey	Generic sports analytics; weak link to aero or anomaly.	Could cite a comprehensive ML survey in autonomous vehicles or motorsport if available.

Recent Reference Recommendations: To strengthen the paper, consider adding recent IEEE/Scopus-indexed works, such as:

- *Ruff et al.*, "A Unifying Review of Deep and Shallow Anomaly Detection"【8†L209-L212】 (IEEE Proc. 2021) – for modern anomaly detection context.
- *Wang et al.*, "Domain Generalization in Time Series Forecasting" (KDD 2022 or similar) – on generalizing time-series models to unseen domains.
- *Yousefi et al.*, any 2023 arXiv or journal article on predictive modeling in automotive/motorsport (if available) or on active aero.
- An updated reference on F1 aerodynamics (e.g. recent SAE or motorsport journals) or on ML in automotive systems (e.g. CNN-RNN hybrid reviews).

PHASE 4: Figures, Tables & Diagrams Review

- **Figure Quality:** Figures (e.g. system architecture, ROC curves, metric bar graphs) are generally clear. Captions are descriptive. However, the resolution in the PDF is somewhat low; labels on axes and legends could be larger or thicker for clarity in final submission. Ensure all text in figures is legible when printed.
- **Readability:** Some plots (ROC curves and bar charts) use similar colors that may not be high contrast. Ensure color choices are distinct and consider texture/hatching for the printed version. For instance, Figure 4's ROC plot lines should have distinguishable styles or colors. Also, axes should be labeled with font sizes consistent with IEEE standards.
- **Labeling & Notation:** Figure captions should explicitly define all acronyms and symbols used. For example, "Phase 1" and "Phase 2" should be explained in captions if not obvious. Tables (III, IV, V, etc.) should have clear headers and units. In Table IV (Experimental Config), some entries (like "Times per epoch") lack context – clarify units or settings.
- **Consistency:** Notation should match text. If "H=10" means 1 second, label it clearly. Check that figures use the same feature names (e.g. "Gear" vs "nGear") as the text. If speed is in km/h, include units.
- **Design Improvements:** For publication quality:
- Redraw the architecture diagram (Figure 1) with vector graphics to ensure crisp lines.
- Add gridlines or minor ticks in performance graphs for better readability.
- Ensure all table fonts and formatting match IEEE style (e.g. centered text in cells, consistent caption placement).
- For metrics bars (Figure 8,10), clearly mark which bar is which model (use legends or direct labels), and consider ordering bars by performance for easier comparison.
- If space allows, merge related figures (e.g. macro vs X-mode F1) to avoid crowding.

PHASE 5: Technical Reviewer Simulation

Reviewer 1 (Technical)

Strengths:

- Addresses a novel and timely problem (2026 F1 active aero) with a well-structured two-phase approach.

- Solid methodological foundation with both classical and deep models, and appropriate statistical analysis (AUC, bootstrap CIs).
- Insightful finding that simple logistic regression outperforms deep networks in a cross-domain setting, which is counter-intuitive and valuable.
- Use of interpretability (Integrated Gradients) to diagnose model behavior is a strong plus.

Weaknesses:

- Limited novelty in ML methodology (mostly known algorithms). The novelty is more in application than in algorithmic innovation.
- The two subtopics (aero mode vs anomaly detection) feel loosely connected; the link could be better justified.
- Some methodological details are missing (see Phase 4).

Major Concerns:

- Justification of K-Means step is unclear: how do the clusters contribute to analysis?
- Lack of a simple rule-based baseline (e.g. threshold on speed/gear) makes it hard to quantify ML gain.
- Domain adaptation issues: the models struggle cross-circuit; perhaps a domain-adaptive approach is needed or at least discussed.

Minor Concerns:

- Typos or formatting glitches (e.g. “contamina tion” split, Eq.5 layout).
- The use of deep models seems overkill given the data size; this should be acknowledged.
- The title and abstract could be more concise.

Questions for Authors:
1. How was the clustering (K=4) in Phase 1 used in the analysis? Why K=4?
2. What drivers/cars are included in each dataset? Does cross-driver variability affect results?
3. How sensitive are results to the choice of horizon H=10? Have you tried H=5 or H=20?

Recommendation: Weak Accept. The paper is technically sound and presents interesting findings, but clarity and completeness can be improved.

Reviewer 2 (Experimental/Results)

Strengths:

- Comprehensive experiments with a range of models and rigorous evaluation (LOCO cross-validation).
- Clear performance reporting (AUC, F1 scores) with confidence intervals.
- The failure case analysis (Monaco GRU) is detailed and adds credibility.

Weaknesses:

- Experimental comparisons miss a trivial baseline (e.g. speed/gear threshold or a majority classifier).

- Only one forecast horizon (1 second) is evaluated; results might differ for shorter/longer horizons.

Major Concerns:

- Overfitting risk: Did you use a validation set from the training circuits to choose hyperparameters? How many epochs were run?
- Statistical testing: You mention significance but should state the method (e.g. bootstrap percentile CI, test statistic).

Minor Concerns:

- Figures could include error bars or indicate variability.
- The "circuit-adaptive metric" is mentioned but not clearly defined or justified experimentally.

Questions for Authors:
1. Can you provide confusion matrices for the best and worst models to complement AUC/F1?
2. How was class imbalance handled besides focal loss? Did you try up/down sampling?
3. Were multiple random splits tried, or is LOCO the only variance source?

Recommendation: Accept. Strong experimental design and significant results, with some room for more baselines and clarity.

Reviewer 3 (Writing/Presentation)

Strengths:

- The paper is generally well-organized with appropriate sectioning.
- Use of tables and figures to summarize results is good. Captions are informative.
- Academic tone and language are mostly professional.

Weaknesses:

- Several grammatical issues and typos exist (e.g. "contamina tion", inconsistent hyphenation).
- Some sentences are lengthy and complex, which hurts readability.
- The abstract and conclusion could be more polished (see Phase 2 suggestions).

Major Concerns:

- Title is too long and somewhat confusing. It should be shortened and focused.
- The related work section could flow better; it reads as a list of references. Integrating the citations into a narrative would improve flow.
- Equation formatting is inconsistent (e.g. spacing).

Minor Concerns:

- Ensure acronyms are defined on first use (e.g. "F1" is obvious, but "GPU" or "AUC" should be defined).
- In-text citation style should be consistent (some places the comma before the citation is missing).
- Some figure/table references in text might be incorrect if numbering changed (verify all).

Questions for Authors:
1. Could you clarify the distinction between X-Mode and Z-Mode in the introduction for readers new to F1?
2. Why was k=4 chosen for clustering? The elbow plot or criteria could be shown.
3. Are all figures and tables referred to in the correct order in the text?

Recommendation: Weak Accept. Interesting and relevant work, but the paper needs careful proofreading and minor reorganization before final publication.

PHASE 6: Acceptance Probability Assessment

- **Novelty (/10):** 6. The application is novel (active-aero forecasting under new F1 rules), but the ML techniques themselves are standard. The two-phase approach is a creative combination, so moderate novelty.
- **Technical Quality (/10):** 8. The methodology is sound and thorough. No major algorithmic innovations, but the implementation is solid.
- **Experimental Strength (/10):** 7. The evaluation is extensive (multiple models, multiple circuits, statistical tests). Missing some baselines hurts a bit, but overall strong.
- **Writing Quality (/10):** 6. Generally clear, but marred by some awkward phrasing, minor errors, and disorganized flow in parts. Could benefit from editing.
- **Citation Quality (/10):** 5. Fair number of relevant refs, but also some weak/outdated ones, and a few missing citations as noted. Needs tightening.
- **Reproducibility (/10):** 7. Most model details are given, but some data split and training details are vague. No code or seed, which lowers score slightly.
- **Overall Conference Readiness (/10):** 7. The paper has strong results and addresses a timely topic, but polishing (clarity, completeness) is needed.

Current Acceptance Probability: ~70%. The core contributions and results are solid for a specialized conference, but weaknesses in writing and some experimental gaps could lead to critiques.

Acceptance Probability After Revisions: ~90%. With careful revision (tightened writing, missing details added, citation fixes), the paper should be well-received by a top-tier venue in this niche domain. The main results are compelling enough to merit acceptance if the criticisms are addressed.

PHASE 7: Publication Improvement Plan

Critical Issues (Must Fix Before Submission)

- **Missing Method Details:** The K-Means clustering step in Phase 1 is not described in the methodology. *Why fix:* Reviewers will be confused about how clustering was used. *Solution:* Add a short explanation (e.g. “We applied K-Means (k=4, chosen by the elbow method) to identify modes of the telemetry features.”). *Impact:* Clarifies a key step, increasing reproducibility and understanding.

- **Baseline Models:** No simple heuristic or constant baseline is reported. *Why:* Without a trivial baseline (e.g. always predict the dominant class or use the known gear-speed rule), it’s hard to gauge the actual improvement. *Solution:* Include at least one naive baseline (e.g. “Predict X-mode when speed≥290km/h & gear≥6”). *Impact:* Strengthens claims that ML adds value, addresses a likely reviewer question.

- **Title/Abstract Clarity:** The title is convoluted and the abstract is too dense. *Why:* First impressions matter; a confusing title/abstract can bias reviewers negatively. *Solution:* Rewrite the title for brevity (e.g. focus on predictive mode detection) and the abstract as suggested above (concise, clear). *Impact:* Improves readability and ensures the paper's novelty is communicated immediately.

High-Priority Improvements

- **LOCO Protocol Explanation:** Explicitly describe how circuits are partitioned (training vs test) in Phase 2. *Why:* Clarifies the experimental design and avoids confusion. *Solution:* Add a sentence like "We perform leave-one-circuit-out cross-validation: each of the four circuits is held out as a test set while training on the other three." *Impact:* Improves transparency of evaluation.

- **Reproducibility Details:** Add information on random seeds, software (library/versions), and hardware. *Why:* IEEE reviewers often expect reproducibility details. *Solution:* In an experimental setup subsection, note "Models were implemented in PyTorch X.X on an NVIDIA YYY GPU. Random seed was set to Z for all runs." *Impact:* Increases confidence in reproducibility.

- **Statistical Reporting:** Report exact confidence intervals or p-values. *Why:* The claim of significance is good, but reviewers will want numbers. *Solution:* Include e.g. "LR outperforms GRU with ΔAUC=0.235 (95% CI [0.12, 0.35], bootstrap)." *Impact:* Strengthens evidence of claims.

- **Writing and Grammar:** Systematically proofread for typos, grammar, and consistency. *Why:* In professional publication, errors undermine credibility. *Solution:* Fix split words ("contamina tion" → "contamination"), ensure acronym definitions (e.g. "LR" at first use), correct any mismatched references/figure numbers. *Impact:* Makes the paper polished and clear.

Medium-Priority Improvements

- **Literature Update:** Replace or remove irrelevant references ([21–23] as above) and cite more relevant recent work. *Why:* Improves scholarly context and avoids reviewer criticism of weak citations. *Solution:* As noted, drop unrelated refs and add domain-adaptation or anomaly detection surveys【6†L51-L59】【8†L209-L212】. *Impact:* Shows the authors did their homework and situates the work properly.

- **Expand Discussion:** Briefly discuss why deep models underperform (e.g. limited data, domain shift) and possibly suggest remedies. *Why:* Anticipates reviewer "why not?" questions and demonstrates insight. *Solution:* Add a paragraph after results: "The poor cross-track performance of recurrent models suggests overfitting to circuit-specific features. Techniques like domain-invariant feature learning【6†L51-L59】 could be explored to address this." *Impact:* Strengthens the analysis and points to future work.

- **Figure Enhancements:** Increase font sizes and contrast in all figures. *Why:* Readability on publication pages is important. *Solution:* Use larger fonts, bold lines in plots. *Impact:* Improves visual quality and professional appearance.

Minor Improvements

- **Table Formatting:** Align numbers and improve caption clarity (e.g. add units). *Why:* Improves presentation quality. *Solution:* Reformat tables with IEEE style (e.g. center labels). *Impact:* Minor polish.
- **Flow and Transitions:** Add a sentence at end of Introduction summarizing paper organization. *Why:* Typical IEEE style. *Solution:* "The rest of this paper is organized as follows: Section II…." *Impact:* Helps reader orientation.
- **Terminology Consistency:** Use consistent naming (e.g., use "X-Mode" vs "xMode" uniformly). *Why:* Avoids confusion. *Solution:* Decide on one style and apply globally. *Impact:* Minor but important detail fix.

PHASE 8: Final Deliverables

1. **Complete Issue List:** All identified problems have been enumerated above in the section-by-section review and improvement plan. They include: missing citations (especially on domain adaptation), incomplete methodology description (K-Means step), absence of a naive baseline, heavy title/abstract phrasing, some outdated/irrelevant references, minor writing errors, and a few gaps in experimental detail (e.g. data splits, hyperparameter tuning, hardware).

2. **Detailed Revision Recommendations:** See *Critical Issues* and *High/Medium/Minor Improvements* above. Each issue is paired with a concrete solution (e.g., add baseline, rewrite title, cite domain adaptation theory【6†L51-L59】).

3. **Suggested Rewritten Text:** For key sections (e.g. Abstract, possibly first paragraph of Introduction), rewritten versions have been given. For example, the *Improved Abstract* is provided above. Likewise, we might suggest rewording any identified awkward sentences (e.g. splitting long sentences in introduction, clarifying transitions).

4. **Missing Experiments:**

5. **Rule-Based Baseline:** Run a simple heuristic (e.g. if speed≥290 km/h and gear≥6 then X-mode; else Z-mode) and report its AUC/F1 as an additional baseline.

6. **Horizon Ablation:** Evaluate at H=5 (0.5s) and H=20 (2s) to see how prediction difficulty scales with lookahead.

7. **Feature Ablation:** Retrain LR without the gear feature to quantify its impact (beyond permutation). This was hinted by permutation importance but a model variant would confirm effect on metrics.

8. **Missing Citations:** Include references to cover the missing items: e.g. Ganin *et al.* (domain adaptation)【6†L51-L59】 in Related Work, and Chandola *et al.* (anomaly detection survey, if needed) or more recent anomaly detection surveys【8†L209-L212】. Possibly cite the FIA 2026 technical regulations or a news article【16†L100-L103】 for active aero background.

9. **Figure Improvement Suggestions:** As noted, use vector graphics, increase text size, and ensure all elements (axes labels, legends) are clear. For example, in Figure 8 and 10, add data labels or error

bars if space allows. Ensure color-blind-friendly palettes. Verify figure numbering and cross-references in text.

10. **Reference Improvement Suggestions:** Remove or replace irrelevant references ([21-23]). Add more up-to-date and on-topic citations (domain adaptation in time-series, automotive anomaly detection). Verify all in-text citations match the reference list after edits.

11. **Reviewer-Style Verdict:** Each of the three simulated reviewers provided summaries above. Briefly, Reviewer 1 (Technical) had mostly minor critiques and was leaning **Weak Accept**. Reviewer 2 (Experimental) found the results convincing and recommended **Accept**. Reviewer 3 (Writing) noted presentation flaws but no fatal errors, recommending **Weak Accept**. Overall, the consensus suggests the paper is on the right track with some concerns to address.

12. **Acceptance Probability:** The estimated acceptance chances are ~70% in its current form, rising to ~90% after revisions. The primary risk is reviewer criticism of presentation and missing details, not the core research quality. Addressing the listed issues should solidify the paper's standing.

13. **Final Recommendation: Accept (pending major revision).** The paper makes a valuable application contribution with thorough experiments and interesting findings. However, to meet IEEE conference standards, the authors should *revise for clarity, completeness, and formatting* as detailed above. The key scientific results are sound; fixing the presentation and methodological details will ensure smooth acceptance.
